diagnosis of exclusion is based largely on history and physical examination and its treatment is based on a minimally invasive algorithm, with the focus on the patient's clinical phenotype and the initial implementation of conservative therapeutic measures, IC/BPS can be well managed even in resource-poor settings. As with many poorly understood and difficult-to-treat conditions, the greatest barrier to its diagnosis and treatment may perhaps be its recognition.

FURTHER READING

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55 Azotemia and Urinary Abnormalities

David B. Mount

Normal kidney functions occur through numerous cellular processes to maintain body homeostasis. Disturbances in any of these functions can lead to abnormalities that may be detrimental to survival. Clinical manifestations of these disorders depend on the pathophysiology of renal injury and often are identified as a complex of symptoms, abnormal physical findings, and laboratory changes that constitute specific syndromes. These renal syndromes (Table 55-1) may arise from systemic illness or as primary renal disease. Nephrologic syndromes usually consist of several elements that reflect the underlying pathologic processes, typically including one or more of the following: (1) reduction in glomerular filtration rate (GFR), (2) abnormalities of urine sediment (red blood cells [RBCs], white blood cells [WBCs], casts, and crystals), (3) abnormal urinary excretion of serum proteins (proteinuria), (4) disturbances in urine volume (oliguria, anuria, polyuria), (5) presence of hypertension and/or expanded total body fluid volume (edema), (6) electrolyte abnormalities, and (7) in some syndromes, fever/pain. The specific combination of these findings should permit identification of one of the major nephrologic syndromes (Table 55-1) and allow differential diagnoses to be narrowed so that the appropriate diagnostic and therapeutic course can be determined. All these syndromes and their associated diseases are discussed in more detail in subsequent chapters. This chapter focuses on several aspects of renal abnormalities that are critically important for distinguishing among those processes: (1) reduction in GFR, (2) alterations of the urinary sediment and/or protein excretion, and (3) abnormalities of urinary volume.

AZOTEMIA

ASSESSMENT OF GFR

Monitoring the GFR is important in both hospital and outpatient settings, and several different methodologies are available. GFR is the primary metric for kidney "function," and its direct measurement involves administration of a radioactive isotope (such as inulin or iothalamate) that is filtered at the glomerulus into the urinary space but is neither reabsorbed nor secreted throughout the tubule. GFR—i.e., the clearance of inulin or iothalamate in milliliters per minute—is calculated from the rate of appearance of the isotope in the urine over several hours. In most clinical circumstances, direct GFR measurement is not feasible, and the plasma creatinine level is used as a surrogate to estimate GFR. Plasma creatinine (P_{Cr}) is the most widely used marker for GFR, which is related directly to urine creatinine (U_{Cr}) excretion and inversely to P_{Cr}. On the basis of this relationship (with some important caveats, as discussed below), GFR will fall in roughly inverse proportion to the rise in P_{Cr}. Failure to account for GFR reductions in drug dosing can lead to significant morbidity and death from drug toxicities (e.g., digoxin, imipenem). In the outpatient setting, P_{Cr} serves as an estimate for GFR (although much less accurate; see below). In patients with chronic progressive renal disease, there is an approximately linear relationship between $1/P_{Cr}$ (*y* axis) and time (*x* axis). The slope of that line will remain constant for an individual; when values deviate, an investigation for a superimposed acute process (e.g., volume depletion, drug reaction) should be initiated. Signs and symptoms of uremia, the clinical symptom complex associated with renal failure, develop at significantly different levels of P_{Cr}, depending on the patient (size, age, and sex), underlying renal disease, existence of concurrent diseases, and true GFR. Generally, patients do not develop symptomatic uremia until renal insufficiency is severe (GFR <15 mL/min).

A significantly reduced GFR (either acute or chronic) is usually reflected in a rise in P_{Cr}, leading to retention of nitrogenous waste products (defined as azotemia) such as urea. Azotemia may result from reduced renal perfusion, intrinsic renal disease, or postrenal processes (ureteral obstruction; see below and Fig. 55-1). Precise determination of GFR is problematic, as both commonly measured indices (urea and creatinine) have characteristics that affect their accuracy as markers of clearance. Urea clearance may underestimate GFR significantly because of urea reabsorption by the tubule. In contrast, creatinine is derived from muscle metabolism of creatine, and its generation varies little from day to day.

Creatinine clearance (CrCl), an approximation of GFR, is measured from plasma and urinary creatinine excretion rates for a defined period (usually 24 h) and is expressed in milliliters per minute: $CrCl = (U_{vol} \times U_{Cr})/(P_{Cr} \times T_{min})$. The "adequacy" or "completeness" of the urinary collection is estimated by the urinary volume and creatinine content; creatinine is produced from muscle and excreted at a relatively constant rate. For a 20- to 50-year-old man, creatinine excretion should be 18.5–25.0 mg/kg body weight; for a woman of the same age, it should be 16.5–22.4 mg/kg body weight. For example, an 80-kg man should excrete between ~1500 and 2000 mg of creatinine in an "adequate" collection. Creatinine is useful for estimating GFR because it is a small, freely filtered solute that is not reabsorbed by the tubules. P_{Cr} levels can increase acutely from dietary ingestion of cooked meat, however, and creatinine can be secreted into the proximal tubule through an organic cation pathway (especially in advanced progressive chronic kidney disease [CKD]), leading to overestimation of GFR. When a timed collection for CrCl is not available, decisions about drug dosing must be based on P_{Cr} alone. Two formulas are used widely to estimate kidney function from P_{Cr}: (1) Cockcroft-Gault and (2) four-variable MDRD (Modification of Diet in Renal Disease).

Cockcroft-Gault:

$$\textbf{CrCl(mL/min)} = \frac{(140-\text{age})\times\text{Lean Body Weight (kg)}}{\text{Serum Creatinine (mg/dL)}\times 72}(\times 0.85 \text{ if female})$$

$$\text{MDRD: eGFR (mL/min per 1.73 m}^2) = 186.3 \times P_{Cr}\,(e^{-1.154}) \times \text{age}\,(e^{-0.203}) \times (0.742 \text{ if female}) \times (1.21 \text{ if black}).$$

Numerous websites are available to assist with these calculations (*www.kidney.org/professionals/kdoqi/gfr_calculator.cfm*). A newer Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) estimated GFR (eGFR), which was developed by pooling several cohorts

TABLE 55-1 Initial Clinical and Laboratory Database for Defining Major Syndromes in Nephrology

SYNDROME	IMPORTANT CLUES TO DIAGNOSIS	COMMON FINDINGS	CHAP(S). DISCUSSING DISEASE-CAUSING SYNDROME
Acute or rapidly progressive renal failure	Anuria	Hypertension, hematuria	321, 326, 328, 331
	Oliguria	Proteinuria, pyuria	
	Documented recent decline in GFR	Casts, edema	
Acute nephritis	Hematuria, RBC casts	Proteinuria	326
	Azotemia, reduced GFR, oliguria	Pyuria	
	Edema, hypertension	Circulatory congestion	
Chronic renal failure	Azotemia for >3 months	Proteinuria, casts	322
	Symptoms or signs of uremia, (late manifestation), casts	Hypocalcemia, hyperphosphatemia, hyperparathyroidism	
	Symptoms or signs of renal osteodystrophy	Polyuria, nocturia	
	Kidneys reduced in size bilaterally	Edema, hypertension	
	Broad casts in urinary sediment	Hyperkalemia, metabolic acidosis	
Nephrotic syndrome	Proteinuria, with >3.5 g/24 h per 1.73 m^2	Casts	326
	Hypoalbuminemia	Lipiduria	
	Edema	Hypercoagulable state	
	Hyperlipidemia		
Asymptomatic urinary abnormalities	Hematuria		326
	Proteinuria (below nephrotic range)		
	Sterile pyuria, casts		
Urinary tract infection/pyelonephritis	Bacteriuria, with >10^5 cfu/mL	Hematuria	140
	Other infectious agent documented in urine	Mild azotemia and reduced GFR	
	Pyuria, leukocyte casts	Mild proteinuria	
	Frequency, urgency	Fever	
	Bladder tenderness, flank tenderness		
Renal tubular defects	Electrolyte disorders	Hematuria	327, 328
	Polyuria, nocturia	"Tubular" proteinuria (<1 g/24 h)	
	Renal calcification	Enuresis	
	Large kidneys	Electrolyte and/or acid-base abnormalities	
	Renal transport defects	Other electrolyte issues, e.g., hypomagnesemia	
Hypertension	Systolic/diastolic hypertension	Proteinuria	288, 329
		Casts	
		Azotemia	
Nephrolithiasis	Previous history of stone passage or removal	Hematuria	330
	Previous history of stone seen by x-ray	Pyuria	
	Renal colic	Frequency, urgency	
Urinary tract obstruction	Azotemia, oliguria, anuria	Hematuria	331
	Polyuria, nocturia, urinary retention	Pyuria	
	Slowing of urinary stream	Enuresis, dysuria	
	Large prostate, large kidneys		
	Flank tenderness, full bladder after voiding		

Abbreviations: cfu, colony-forming units; GFR; glomerular filtration rate; RBC, red blood cell.

with and without kidney disease who had data on directly measured GFR, appears to be more accurate:

$$\text{CKD-EPI: eGFR} = 141 \times \min(P_{Cr}/k, 1)^{a} \times \max(P_{Cr}/k, 1)^{-1.209} \times 0.993^{Age} \times 1.018 \text{ [if female]} \times 1.159 \text{ [if black]},$$

where P_{Cr} is plasma creatinine, k is 0.7 for females and 0.9 for males, a is −0.329 for females and −0.411 for males, *min* indicates the minimum of P_{Cr}/k or 1, and *max* indicates the maximum of P_{Cr}/k or 1 (*https://www.mdcalc.com/ckd-epi-equations-glomerular-filtration-rate-gfr*).

There are limitations to all creatinine-based estimates of GFR. Each equation, along with 24-h urine collection for measurement of creatinine clearance, is based on the assumption that the patient is in *steady state*, without daily increases or decreases in P_{Cr} as a result of rapidly changing GFR. The MDRD equation is better correlated with true GFR when the GFR is <60 mL/min per 1.73 m^2. The gradual loss of muscle from chronic illness, chronic use of glucocorticoids, or malnutrition can mask significant changes in GFR with small or imperceptible changes in P_{Cr}.

The coefficient of 1.159 in the CKD-EPI equation to adjust for self-reported black race reflects that measured GFR was 16% higher in blacks than nonblacks with similar age, sex, and creatinine in the data set used to develop the equation. Race is a social rather than a biological construct, for which reason the use of the "race modifier" in calculating eGFR using CKD-EPI and other equations has come under scrutiny. In particular, given the implications of utilizing self-reported race to modify clinical laboratory results, many medical centers have recently stopped reporting eGFRs that have been calculated using a race modifier. This change is projected to have positive consequences, in particular, improved access to waitlisting for renal transplantation in black patients at an earlier stage of CKD. Potential

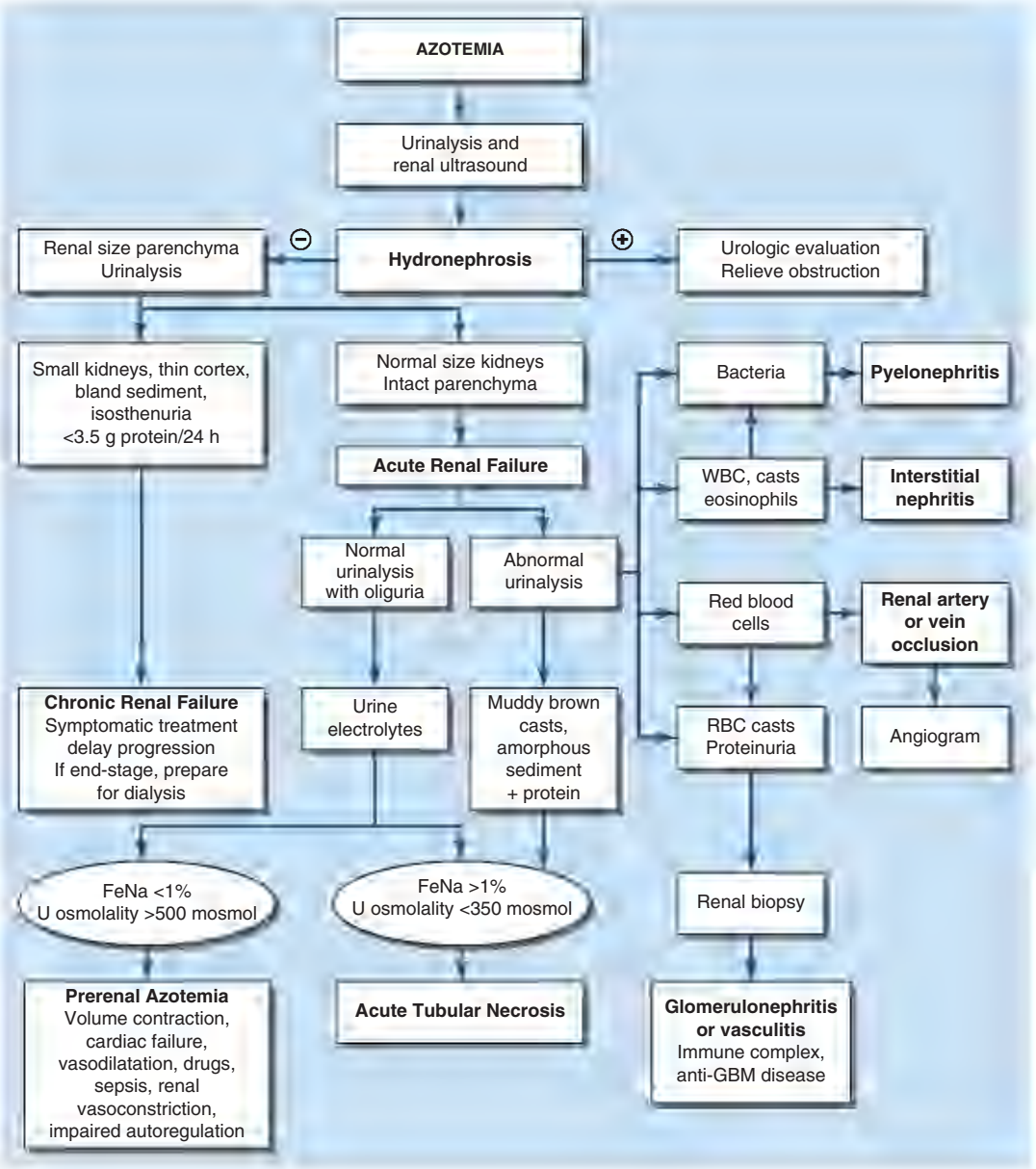

FIGURE 55-1 Approach to the patient with azotemia. FeNa, fractional excretion of sodium; GBM, glomerular basement membrane; RBC, red blood cell; U, urine; WBC, white blood cell.

negative consequences include "overdiagnosis" of CKD, inadequate or inaccurate dosing of drugs that are eliminated through the kidney (e.g., metformin), reduced access to imaging modalities for black patients with CKD with a lower reported eGFR, and reductions in living kidney donation among blacks. These and the other limitations in creatinine-based eGFR have led to the development of alternative methods for estimating GFR.

Cystatin C, a member of the cystatin superfamily of cysteine protease inhibitors, is produced at a relatively constant rate from all nucleated cells. Serum cystatin C has been proposed to be a more sensitive marker of early GFR decline than is P_{Cr}, with lesser effects of muscle mass on circulating levels; however, cystatin C levels are influenced by the patient's sex and the presence of diabetes mellitus, smoking, and inflammation. To the extent that cystatin C–based calculation of eGFR is less affected by self-reported race and muscle mass, it is an increasingly important adjunct to creatinine-based eGFR. Recently, eGFR equations that include *both* creatinine and cystatin C have been shown to be more accurate than the single-measurement equations. Clinical judgement and clinical assessment also play an important role in interpreting eGFR values. For example, a bodybuilder may have an elevated creatinine level due to increased muscle mass, with an underestimate of GFR based on a creatinine-based eGFR; in that case, the cystatin C eGFR may be more accurate.

APPROACH TO THE PATIENT

Azotemia

Once GFR reduction has been established, the physician must decide if it represents acute or chronic renal injury. The clinical circumstances, history, and laboratory data often make this an easy distinction. However, the laboratory abnormalities characteristic of chronic renal failure, including anemia, hypocalcemia, and hyperphosphatemia, are also often present in patients presenting with acute renal failure. Radiographic evidence of renal osteodystrophy (Chap. 322) can be seen only in chronic renal failure but is a very late finding, typically in patients with end-stage renal disease (ESRD) maintained on dialysis. The urinalysis and renal ultrasound can facilitate distinguishing acute from chronic renal failure. An approach to the evaluation of azotemic patients is shown in Fig. 55-1. Patients with advanced chronic renal insufficiency often have some proteinuria, nonconcentrated urine (isosthenuria; isosmotic with plasma),

and small kidneys on ultrasound, characterized by increased echogenicity and cortical thinning. Treatment should be directed toward slowing the progression of renal disease and providing symptomatic relief for edema, acidosis, anemia, and hyperphosphatemia, as discussed in **Chap. 322**. Acute renal failure **(Chap. 321)** can result from processes that affect blood flow and glomerular perfusion (prerenal azotemia), intrinsic renal diseases (affecting small vessels, glomeruli, or tubules), or postrenal processes (obstruction of urine flow in ureters, bladder, or urethra) **(Chap. 331)**.

PRERENAL FAILURE

Decreased renal perfusion accounts for 40–80% of cases of acute renal failure and, if appropriately treated, is readily reversible. The etiologies of prerenal azotemia include any cause of decreased circulating blood volume (gastrointestinal hemorrhage, burns, diarrhea, diuretics), volume sequestration (pancreatitis, peritonitis, rhabdomyolysis), or decreased effective arterial volume (cardiogenic shock, sepsis). Renal and glomerular perfusion also can be affected by reductions in cardiac output from peripheral vasodilation (sepsis, drugs) or profound renal vasoconstriction (severe heart failure, hepatorenal syndrome, agents such as nonsteroidal anti-inflammatory drugs [NSAIDs]). True or "effective" arterial hypovolemia leads to a fall in mean arterial pressure, which in turn triggers a series of neural and humoral responses, including activation of the sympathetic nervous and renin-angiotensin-aldosterone systems and vasopressin (AVP) release. GFR is maintained by prostaglandin-mediated dilatation of afferent arterioles and angiotensin II–mediated constriction of efferent arterioles. Once the mean arterial pressure falls below 80 mmHg, GFR declines steeply.

Blockade of prostaglandin production by NSAIDs can result in severe vasoconstriction and acute renal failure. Blocking angiotensin action with angiotensin-converting enzyme (ACE) inhibitors or angiotensin receptor blockers (ARBs) decreases efferent arteriolar tone and in turn decreases glomerular capillary perfusion pressure. Patients taking NSAIDs and/or ACE inhibitors/ARBs are most susceptible to hemodynamically mediated acute renal failure when blood volume or arterial perfusion pressure is reduced for any reason; under these circumstances, preservation of GFR is dependent on afferent vasodilation due to prostaglandins and efferent vasoconstriction due to angiotensin II. Patients with bilateral renal artery stenosis (or stenosis in a solitary kidney) can also be dependent on efferent arteriolar vasoconstriction for maintenance of glomerular filtration pressure and are particularly susceptible to a precipitous decline in GFR when given ACE inhibitors or ARBs.

Prolonged renal hypoperfusion may lead to acute tubular necrosis (ATN), an intrinsic renal disease that is discussed below. The urinalysis and urinary electrolyte measurements can be useful in distinguishing prerenal azotemia from ATN **(Table 55-2)**. The urine Na and osmolality of patients with prerenal azotemia can be predicted from the stimulatory actions of norepinephrine, angiotensin II, AVP, aldosterone, and low tubule fluid flow rate. In prerenal conditions, the tubules are intact, leading to a concentrated urine (>500 mosmol), avid Na retention (urine Na concentration, <20 mmol/L; fractional excretion of Na [FE_{Na}], <1%), and U_{Cr}/P_{Cr} >40 (Table 55-2). The FE_{Na} is typically >1% in ATN but may be <1% in patients with milder, nonoliguric ATN (e.g., from rhabdomyolysis) and in patients with underlying "prerenal" disorders, such as congestive heart failure (CHF) or cirrhosis or hepatorenal syndrome. The prerenal urine sediment is usually normal or has hyaline and granular casts, whereas the sediment of ATN usually is filled with cellular debris, tubular epithelial casts, and dark (muddy brown) granular casts. Microscopic examination of a urine sediment is a key test in AKI, since the presence of dark granular casts and/or tubular epithelial cells in the urine is highly predictive of ATN. The measurement of urinary biomarkers associated with tubular injury is a promising technique to detect subclinical ATN and/or help further diagnose the exact cause of acute renal failure.

TABLE 55-2 Laboratory Findings in Acute Renal Failure

INDEX	PRERENAL AZOTEMIA	OLIGURIC ACUTE RENAL FAILURE
BUN/P_{Cr} ratio	>20:1	10–15:1
Urine sodium U_{Na}, meq/L	<20	>40
Urine osmolality, mosmol/L H_2O	>500	<350
Fractional excretion of sodium[a]	<1%	>2%
Urine/plasma creatinine U_{Cr}/P_{Cr}	>40	<20
Urinalysis (casts)	None or hyaline/granular	Muddy brown

$$^{a}FE_{Na} = \frac{U_{Na} \times P_{Cr} \times 100}{P_{Na} \times U_{cr}}$$

Abbreviations: BUN, blood urea nitrogen; P_{Cr}, plasma creatinine concentration; P_{Na}, plasma sodium concentration; U_{Cr}, urine creatinine concentration; U_{Na}, urine sodium concentration.

POSTRENAL AZOTEMIA

Urinary tract obstruction accounts for <5% of cases of acute renal failure but is usually reversible and must be ruled out early in the evaluation (Fig. 55-1). Since a single kidney is capable of adequate clearance, complete obstructive acute renal failure requires obstruction at the urethra or bladder outlet, bilateral ureteral obstruction, or unilateral obstruction in a patient with a single functioning kidney. Obstruction is usually diagnosed by the presence of ureteral and renal pelvic dilation on renal ultrasound. However, early in the course of obstruction or if the ureters are unable to dilate (e.g., encasement by pelvic or periureteral tumors or by retroperitoneal fibrosis), the ultrasound examination may be negative. Other imaging, such as a furosemide renogram (MAG3 nuclear medicine study), may be required to better define the presence or absence of obstructive uropathy. The specific urologic conditions that cause obstruction are discussed in **Chap. 331**.

INTRINSIC RENAL DISEASE

When prerenal and postrenal azotemia have been excluded as etiologies of renal failure, an intrinsic parenchymal renal disease is present. Intrinsic renal disease can arise from processes involving large renal vessels, intrarenal microvasculature and glomeruli, or the tubulointerstitium. Ischemic and toxic ATN account for ~90% of cases of acute intrinsic renal failure. As outlined in Fig. 55-1, the clinical setting and urinalysis are helpful in separating the possible etiologies. Prerenal azotemia and ATN are part of a spectrum of renal hypoperfusion; evidence of structural tubule injury is present in ATN, whereas prompt reversibility occurs with prerenal azotemia upon restoration of adequate renal perfusion. Thus, ATN often can be distinguished from prerenal azotemia by urinalysis and urine electrolyte composition (Table 55-2 and Fig. 55-1). Ischemic ATN is observed most frequently in patients who have undergone major surgery, trauma, severe hypovolemia, overwhelming sepsis, or extensive burns. Nephrotoxic ATN complicates the administration of many common medications, usually by inducing a combination of intrarenal vasoconstriction, direct tubule toxicity, and/or tubular obstruction. The kidney is vulnerable to toxic injury by virtue of its rich blood supply (25% of cardiac output) and its ability to concentrate and metabolize toxins. A diligent search for hypotension and nephrotoxins usually uncovers the specific etiology of ATN. Discontinuation of nephrotoxins and stabilization of blood pressure often suffice without the need for dialysis, with ongoing regeneration of tubular cells. **An extensive list of potential drugs and toxins implicated in ATN is found in Chap. 321.**

Processes involving the tubules and interstitium can lead to acute kidney injury (AKI), a subtype of acute renal failure. These processes include drug-induced interstitial nephritis (especially by antibiotics, NSAIDs, and proton pump inhibitors), severe infections

(both bacterial and viral), systemic diseases (e.g., systemic lupus erythematosus), and systemic disorders (e.g., sarcoidosis, Sjögren's syndrome, lymphoma, or leukemia). A list of drugs associated with allergic interstitial nephritis is found in **Chap. 328**. Urinalysis usually shows mild to moderate proteinuria, hematuria, and pyuria (~75% of cases) and occasionally WBC casts. The finding of RBC casts in interstitial nephritis has been reported but should prompt a search for glomerular diseases (Fig. 55-1). Renal biopsy will be needed to distinguish among these possibilities. The classic sediment finding in allergic interstitial nephritis is a predominance (>10%) of urinary eosinophils with Wright's or Hansel's stain; however, urinary eosinophils can be increased in several other causes of AKI, such that measurement of urine eosinophils has no diagnostic utility in renal disease. This test is no longer recommended in the workup of AKI.

Occlusion of large renal vessels, including arteries and veins, is an uncommon cause of acute renal failure. A significant reduction in GFR by this mechanism suggests bilateral processes or, in a patient with a single functioning kidney, a unilateral process. In patients with preexisting renal artery stenosis, a substantial renal collateral circulation can develop over time and sustain renal perfusion—typically not enough to sustain glomerular filtration, but enough to maintain tissue viability—in the event of total renal artery occlusion. Renal arteries can be occluded with atheroemboli, thromboemboli, in situ thrombosis, aortic dissection, or vasculitis. Atheroembolic renal failure can occur spontaneously but most often is associated with recent aortic instrumentation. The emboli are cholesterol-rich and lodge in medium and small renal arteries, with a consequent eosinophil-rich inflammatory reaction. Patients with atheroembolic acute renal failure often have a normal urinalysis, but the urine may contain eosinophils and casts. The diagnosis can be confirmed by renal biopsy, but this procedure is often unnecessary when other stigmata of atheroemboli are present (livedo reticularis, distal peripheral infarcts, eosinophilia). Renal artery thrombosis may lead to mild proteinuria and hematuria, whereas renal vein thrombosis typically occurs in the context of heavy proteinuria and hematuria. These vascular complications often require angiography for confirmation and are discussed in **Chap. 329**.

Diseases of the glomeruli (glomerulonephritis and vasculitis) and the renal microvasculature (hemolytic-uremic syndromes, thrombotic thrombocytopenic purpura, and malignant hypertension) usually present with various combinations of glomerular injury: proteinuria, hematuria, reduced GFR, and alterations of sodium excretion that lead to hypertension, edema, and circulatory congestion (acute nephritic syndrome). These findings may occur as primary renal diseases or as renal manifestations of systemic diseases. The clinical setting and other laboratory data help distinguish primary renal diseases from systemic diseases. The finding of RBC casts in the urine is an indication for early renal biopsy (Fig. 55-1), as the pathologic pattern has important implications for diagnosis, prognosis, and treatment. Hematuria without RBC casts can also be an indication of glomerular disease, since RBC casts are highly specific but very insensitive for glomerulonephritis. The specificity of urine microscopy can be enhanced by examining urine with a phase contrast microscope capable of detecting dysmorphic red cells ("acanthocytes") that are associated with glomerular disease. This evaluation is summarized in **Fig. 55-2**. A detailed discussion of glomerulonephritis and diseases of the microvasculature is found in **Chap. 328**.

OLIGURIA AND ANURIA

Oliguria refers to a 24-h urine output <400 mL, and *anuria* is the complete absence of urine formation (<100 mL). Anuria can be caused by complete bilateral urinary tract obstruction; a vascular catastrophe (dissection or arterial occlusion); renal vein thrombosis; acute cast nephropathy in myeloma; renal cortical necrosis; severe ATN; severe rapidly progressive glomerulonephritis; combined therapy with NSAIDs, ACE inhibitors, and/or ARBs; and hypovolemic, cardiogenic, or septic shock. Oliguria is never normal, since at least 400 mL of maximally concentrated urine must be produced to excrete the obligate daily osmolar load. *Nonoliguria* refers to urine output >400 mL/d in patients with acute or chronic azotemia. With nonoliguric ATN, disturbances of potassium and hydrogen balance are less severe than in oliguric patients, and recovery to normal renal function is usually more rapid.

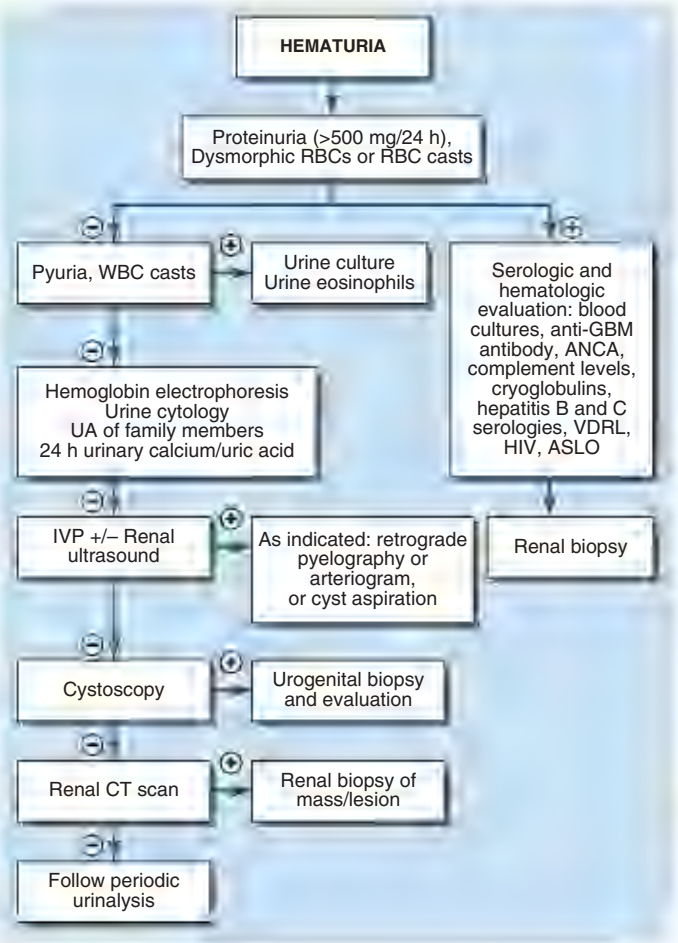

FIGURE 55-2 Approach to the patient with hematuria. ANCA, antineutrophil cytoplasmic antibody; ASLO, antistreptolysin O; CT, computed tomography; GBM, glomerular basement membrane; IVP, intravenous pyelography; RBC, red blood cell; UA, urinalysis; VDRL, Venereal Disease Research Laboratory; WBC, white blood cell.

ABNORMALITIES OF THE URINE

PROTEINURIA

The evaluation of proteinuria is shown schematically in **Fig. 55-3** and typically is initiated after detection of proteinuria by dipstick examination. The dipstick measurement detects only albumin and gives false-positive results at pH >7.0 or when the urine is very concentrated or contaminated with blood. Because the dipstick relies on urinary albumin concentration, a very dilute urine may obscure significant proteinuria on dipstick examination. Quantification of urinary albumin on a spot urine sample (ideally from a first morning void) by measurement of an albumin-to-creatinine ratio (ACR) is helpful in approximating a 24-h albumin excretion rate (AER), where ACR (mg/g) ≈ AER (mg/24 h). Furthermore, proteinuria that is not predominantly due to albumin will be missed by dipstick screening. This information is particularly important for the detection of Bence-Jones proteins in the urine of patients with multiple myeloma. Tests to measure total urine protein concentration accurately rely on precipitation with sulfosalicylic or trichloroacetic acid (Fig. 55-3). As with albuminuria, the ratio of

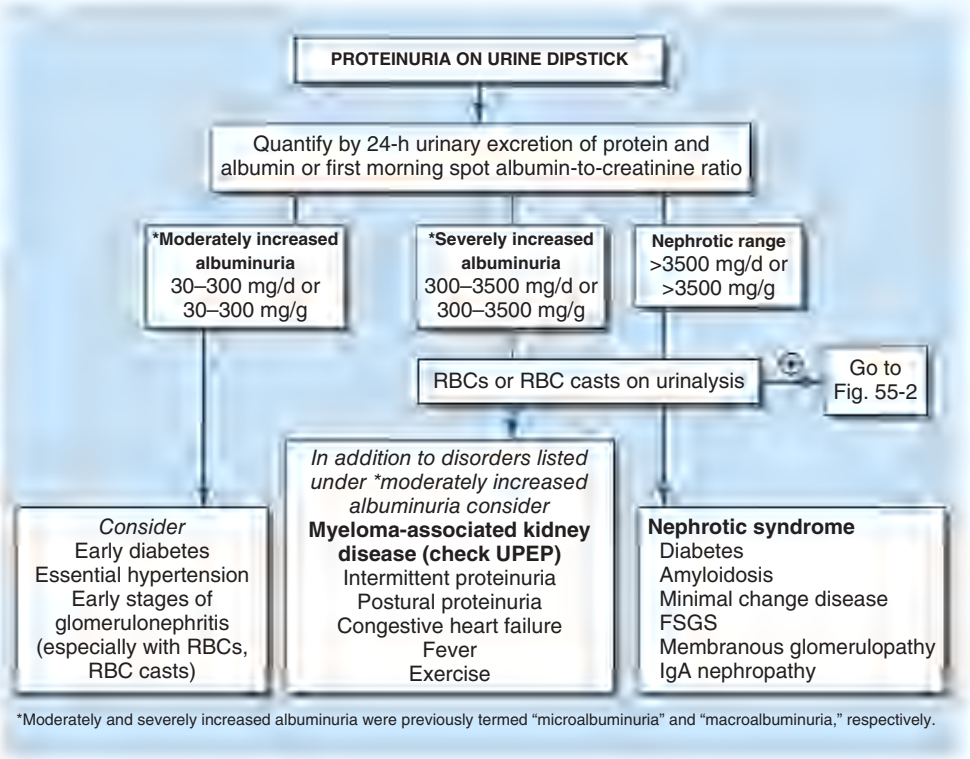

FIGURE 55-3 Approach to the patient with proteinuria. Investigation of proteinuria is often initiated by a positive dipstick on routine urinalysis. Conventional dipsticks detect predominantly albumin and provide a semiquantitative assessment (trace, 1+, 2+, or 3+), which is influenced by urinary concentration as reflected by urine specific gravity (minimum, <1.005; maximum, 1.030). However, more exact determination of proteinuria should employ a spot morning protein/creatinine ratio (mg/g) or a 24-h urine collection (mg/24 h). FSGS, focal segmental glomerulosclerosis; RBC, red blood cell; UPEP, urine protein electrophoresis.

protein to creatinine in a random "spot" urine can also provide a rough estimate of protein excretion; for example, a protein/creatinine ratio of 3.0 correlates to ~3.0 g of proteinuria per day. Formal assessment of urinary protein excretion requires a 24-h urine protein collection (see "Assessment of GFR," above).

The magnitude of proteinuria and its composition in the urine depend on the mechanism of renal injury that leads to protein losses. Both charge and size selectivity normally prevent virtually all plasma albumin, globulins, and other high-molecular-weight proteins from crossing the glomerular wall; however, if this barrier is disrupted, plasma proteins may leak into the urine (glomerular proteinuria; Fig. 55-3). Smaller proteins (<20 kDa) are freely filtered but are readily reabsorbed by the proximal tubule. Typically, healthy individuals excrete <150 mg/d of total protein and <30 mg/d of albumin. However, even at albuminuria levels <30 mg/d, risk for progression to overt nephropathy or subsequent cardiovascular disease is increased. The remainder of the protein in the urine is secreted by the tubules (Tamm-Horsfall, IgA, and urokinase) or represents small amounts of filtered β_2-microglobulin, apoproteins, enzymes, and peptide hormones. Another mechanism of proteinuria entails excessive production of an abnormal protein that exceeds the capacity of the tubule for reabsorption. This situation most commonly occurs with plasma cell dyscrasias, such as multiple myeloma, amyloidosis, and lymphomas, that are associated with monoclonal production of immunoglobulin light chains. Other causes include lysozyme-associated nephropathy, a rare cause of kidney injury in patients with chronic myelomonocytic leukemia (CMML); overproduction of lysozyme results in excessive reabsorption of the enzyme by the proximal tubule, resulting in a severe tubulopathy with intracytoplasmic, membrane-bound vacuoles containing homogenous or granular electron dense material on electron microscopy.

The normal glomerular endothelial cell forms a barrier composed of pores of ~100 nm that retain blood cells but offer little impediment to passage of most proteins. The glomerular basement membrane traps most large proteins (>100 kDa), and the foot processes of epithelial cells (podocytes) cover the urinary side of the glomerular basement membrane and produce a series of narrow channels (slit diaphragms) to allow molecular passage of small solutes and water but not proteins. Some glomerular diseases, such as minimal change disease, cause fusion of glomerular epithelial cell foot processes, resulting in predominantly "selective" (Fig. 55-3) loss of albumin. Other glomerular diseases can present with disruption of the basement membrane and slit diaphragms (e.g., by immune complex deposition), resulting in losses of albumin and other plasma proteins. The fusion of foot processes causes increased pressure across the capillary basement membrane, resulting in areas with larger pore sizes (and more severe "nonselective" proteinuria) (Fig. 55-3).

When the total daily urinary excretion of protein is >3.5 g, hypoalbuminemia, hyperlipidemia, and edema (nephrotic syndrome; Fig. 55-3) are often present as well. However, total daily urinary protein excretion >3.5 g can occur without the other features of the nephrotic syndrome in a variety of other renal diseases, including diabetes (Fig. 55-3). Plasma cell dyscrasias (multiple myeloma) can be associated with large amounts of excreted light chains in the urine, which may not be detected by dipstick. The light chains are filtered by the glomerulus and overwhelm the reabsorptive capacity of the proximal tubule. Renal failure from these disorders occurs through a variety of mechanisms, including but not limited to proximal tubule injury, tubule obstruction (cast nephropathy), amyloid deposition, and light chain deposition (**Chap. 328**). The specific renal lesion is dictated by the sequence and structural characteristics of the monoclonal light chain; however, not all excreted light chains are nephrotoxic.

Hypoalbuminemia in nephrotic syndrome occurs through excessive urinary losses and increased proximal tubule catabolism of filtered albumin. Edema results from renal sodium retention and reduced plasma oncotic pressure, which favors fluid movement from capillaries to interstitium. To compensate for the perceived decrease in effective intravascular volume, activation of the renin-angiotensin system, stimulation of AVP, and activation of the sympathetic nervous system take place, promoting continued renal salt and water reabsorption and progressive edema. Filtered proteases, normally retained by the glomerular filtration barrier, can also directly activate sodium reabsorption by the epithelial Na channels in principal cells (ENaC) in nephrotic syndrome. Despite these changes, hypertension is uncommon in primary kidney diseases resulting in the nephrotic syndrome (Fig. 55-3 and **Chap. 326**). The urinary loss of regulatory proteins and changes in hepatic synthesis contribute to the other manifestations of the nephrotic syndrome. A hypercoagulable state may arise from urinary losses of antithrombin III, reduced serum levels of proteins S and C, hyperfibrinogenemia, and enhanced platelet aggregation. Hypercholesterolemia may be severe and results from increased hepatic lipoprotein synthesis. Loss of immunoglobulins contributes to an increased risk of infection. Many diseases (some listed in Fig. 55-3) and drugs can cause the nephrotic syndrome; a complete list is found in **Chap. 326**.

HEMATURIA, PYURIA, AND CASTS

Isolated hematuria without proteinuria, other cells, or casts is often indicative of bleeding from the urinary tract. Hematuria is defined as two to five RBCs per high-power field (HPF) and can be detected by dipstick. A false-positive dipstick for hematuria (where no RBCs are seen on urine microscopy) may occur when myoglobinuria is present,

often in the setting of rhabdomyolysis. Common causes of isolated hematuria include stones, neoplasms, tuberculosis, trauma, and prostatitis. Gross hematuria with blood clots usually is not an intrinsic renal process; rather, it suggests a postrenal source in the urinary collecting system. Evaluation of patients presenting with microscopic hematuria is outlined in Fig. 55-2. A single urinalysis with hematuria is common and can result from menstruation, viral illness, allergy, exercise, or mild trauma. Persistent or significant hematuria (>3 RBCs/HPF on three urinalyses, a single urinalysis with >100 RBCs, or gross hematuria) is associated with significant renal or urologic lesions in 9.1% of cases. The level of suspicion for urogenital neoplasms in patients with isolated painless hematuria and nondysmorphic RBCs increases with age. Neoplasms are rare in the pediatric population, and isolated hematuria is more likely to be "idiopathic" or associated with a congenital anomaly. Hematuria with pyuria and bacteriuria is typical of infection and should be treated with antibiotics after appropriate cultures. Acute cystitis or urethritis in women can cause gross hematuria. Hypercalciuria and hyperuricosuria are also risk factors for unexplained isolated hematuria in both children and adults. In some of these patients (50–60%), reducing calcium and uric acid excretion through dietary interventions can eliminate the microscopic hematuria.

Isolated microscopic hematuria can be a manifestation of glomerular diseases. The RBCs of glomerular origin are often dysmorphic when examined by phase-contrast microscopy. Irregular shapes of RBCs may also result from pH and osmolarity changes produced along the distal nephron. Observer variability in detecting dysmorphic RBCs is common. The most common etiologies of isolated glomerular hematuria are IgA nephropathy, hereditary nephritis, and thin basement membrane disease. IgA nephropathy and hereditary nephritis can lead to episodic gross hematuria. A family history of renal failure is often present in hereditary nephritis, and patients with thin basement membrane disease often have family members with microscopic hematuria. A renal biopsy is needed for the definitive diagnosis of these disorders, which are discussed in more detail in **Chap. 326**. Hematuria with dysmorphic RBCs, RBC casts, and protein excretion >500 mg/d is virtually diagnostic of glomerulonephritis. RBC casts form as RBCs that enter the tubule fluid and become trapped in a cylindrical mold of gelled Tamm-Horsfall protein. Even in the absence of azotemia, these patients should undergo serologic evaluation and renal biopsy as outlined in Fig. 55-2.

Isolated pyuria is unusual since inflammatory reactions in the kidney or collecting system also are associated with hematuria. The presence of bacteria suggests infection, and WBC casts with bacteria are indicative of pyelonephritis; "sterile pyuria" with negative urinary bacterial cultures can be seen in urogenital tuberculosis. WBCs and/or WBC casts also may be seen in acute glomerulonephritis as well as in tubulointerstitial processes such as interstitial nephritis and transplant rejection.

Casts can be seen in chronic renal diseases. Degenerated cellular casts called *waxy casts* or *broad casts* (arising in the dilated tubules that have undergone compensatory hypertrophy in response to reduced renal mass) may be seen in the urine.

ABNORMALITIES OF URINE VOLUME

■ POLYURIA

By history, it is often difficult for patients to distinguish urinary frequency (often of small volumes) from true polyuria (>3 L/d), and a quantification of volume by 24-h urine collection may be needed **(Fig. 55-4)**. Polyuria results from two potential mechanisms: (1) excretion of nonabsorbable solutes (such as glucose) or (2) excretion of water (usually from a defect in AVP production or renal responsiveness). To distinguish a solute diuresis from a water diuresis and to determine whether the diuresis is appropriate for the clinical circumstances, urine osmolality is measured. The average person excretes between 600 and 800 mosmol of solutes per day, primarily as urea and electrolytes. If the urine output is >3 L/d and the urine is dilute (<250 mosmol/L), total osmolar excretion is normal and a water diuresis is present. This circumstance could arise from polydipsia, inadequate secretion of AVP

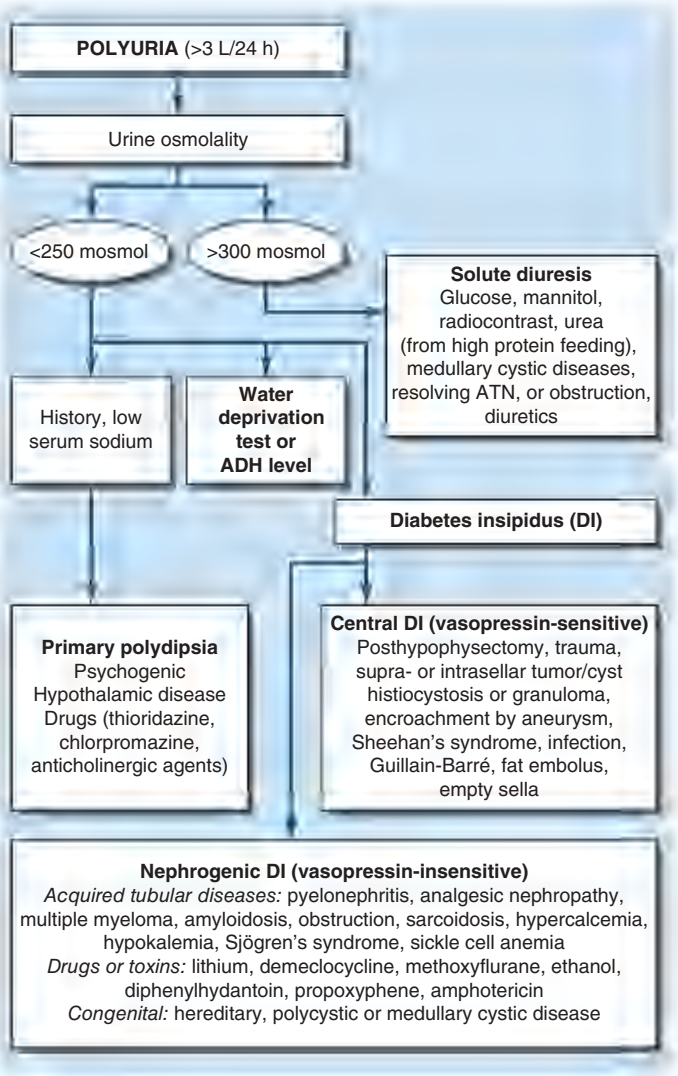

FIGURE 55-4 Approach to the patient with polyuria. ADH, antidiuretic hormone; ATN, acute tubular necrosis.

(*central diabetes insipidus*), or failure of renal tubules to respond to AVP (*nephrogenic diabetes insipidus*). If the urine volume is >3 L/d and urine osmolality is >300 mosmol/L, a solute diuresis is clearly present and a search for the responsible solute(s) is mandatory.

Excessive filtration of a poorly reabsorbed solute such as glucose or mannitol can depress reabsorption of NaCl and water in the proximal tubule and lead to enhanced excretion in the urine. Poorly controlled diabetes mellitus with glucosuria is the most common cause of a solute diuresis, leading to volume depletion and serum hypertonicity. Since the urine sodium concentration is less than that of blood, more water than sodium is lost, causing hypernatremia and hypertonicity. Common iatrogenic solute diuresis occurs in association with mannitol administration, radiocontrast media, and high-protein feedings (enteral or parenteral), leading to increased urea production and excretion. Less commonly, excessive sodium loss may result from cystic renal diseases or Bartter's syndrome or may develop during a tubulointerstitial process (such as resolving ATN). In these so-called salt-wasting disorders, the tubule damage results in direct impairment of sodium reabsorption and indirectly reduces the responsiveness of the tubule to aldosterone. Usually, the sodium losses are mild, and the obligatory urine output is <2 L/d; resolving ATN and postobstructive diuresis are exceptions and may be associated with significant natriuresis and polyuria.

Formation of large volumes of dilute urine is usually due to polydipsic states or diabetes insipidus. Primary polydipsia can result from habit, psychiatric disorders, neurologic lesions, or medications. During

deliberate polydipsia, extracellular fluid volume is normal or expanded and plasma AVP levels are reduced because serum osmolality tends to be near the lower limits of normal. Urine osmolality is also maximally dilute at 50 mosmol/L.

Central diabetes insipidus may be idiopathic in origin or secondary to a variety of conditions, including hypophysectomy, trauma, neoplastic, inflammatory, vascular, or infectious hypothalamic diseases. Idiopathic central diabetes insipidus is associated with selective destruction of the AVP-secreting neurons in the supraoptic and paraventricular nuclei and can either be inherited as an autosomal dominant trait or occur spontaneously. Nephrogenic diabetes insipidus can occur in a variety of clinical situations, as summarized in Fig. 55-4.

A plasma AVP level is recommended as the best method for distinguishing between central and nephrogenic diabetes insipidus. Assays for circulating copeptin, a peptide that is cleaved from pre-pro-AVP during axonal transport in the posterior pituitary, are also now available in many centers. A water deprivation test plus exogenous desmopressin (DDAVP) may distinguish primary polydipsia from central and nephrogenic diabetes insipidus. Measurement of hypertonic saline–stimulated plasma copeptin, if available, can substitute for water deprivation testing. **For a detailed discussion, see Chap. 393.**

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Julie Lin and Brad Denker contributed to this chapter in the 19th edition and some material from that chapter has been retained here.

FURTHER READING

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56 Fluid and Electrolyte Disturbances

David B. Mount

SODIUM AND WATER

COMPOSITION OF BODY FLUIDS

Water is the most abundant constituent in the body, comprising ~50% of body weight in women and 60% in men. Total-body water is distributed in two major compartments: 55–75% is intracellular (intracellular fluid [ICF]), and 25–45% is extracellular (extracellular fluid [ECF]). The ECF is further subdivided into intravascular (plasma water) and extravascular (interstitial) spaces in a ratio of 1:3. Fluid movement between the intravascular and interstitial spaces occurs across the capillary wall and is determined by Starling forces, i.e., capillary hydraulic pressure and colloid osmotic pressure. The transcapillary hydraulic pressure gradient exceeds the corresponding oncotic pressure gradient, thereby favoring the movement of plasma ultrafiltrate into the extravascular space. The return of fluid into the intravascular compartment occurs via lymphatic flow.

The solute or particle concentration of a fluid is known as its osmolality, expressed as milliosmoles per kilogram of water (mOsm/kg). Water easily diffuses across most cell membranes to achieve osmotic equilibrium (ECF osmolality = ICF osmolality). Notably, the extracellular and intracellular solute compositions differ considerably owing to the activity of various transporters, channels, and ATP-driven membrane pumps. The major ECF particles are Na^+ and its accompanying anions Cl^- and HCO_3^-, whereas K^+ and organic phosphate esters (ATP, creatine phosphate, and phospholipids) are the predominant ICF osmoles. Solutes that are restricted to the ECF or the ICF determine the "tonicity" or effective osmolality of that compartment. Certain solutes, particularly urea, do not contribute to water shifts across most membranes and are thus known as *ineffective osmoles*.

Water Balance Vasopressin secretion, water ingestion, and renal water transport collaborate to maintain human body fluid osmolality between 280 and 295 mOsm/kg. Vasopressin (AVP) is synthesized in magnocellular neurons within the hypothalamus; the distal axons of these neurons project to the posterior pituitary or neurohypophysis, from which AVP is released into the circulation. A network of central "osmoreceptor" neurons, which includes the AVP-expressing magnocellular neurons themselves, sense circulating osmolality via nonselective, stretch-activated cation channels. These osmoreceptor neurons are activated or inhibited by modest increases and decreases in circulating osmolality, respectively; activation leads to AVP release and thirst.

AVP secretion is stimulated as systemic osmolality increases above a threshold level of ~285 mOsm/kg, above which there is a linear relationship between osmolality and circulating AVP **(Fig. 56-1)**. Thirst and thus water ingestion are also activated at ~285 mOsm/kg, beyond which there is an equivalent linear increase in the perceived intensity of thirst as a function of circulating osmolality. Changes in blood volume and blood pressure are also direct stimuli for AVP release and thirst, albeit with a less sensitive response profile. Of perhaps greater clinical relevance to the pathophysiology of water homeostasis, ECF volume strongly modulates the relationship between circulating osmolality and AVP release, such that hypovolemia reduces the osmotic threshold and increases the slope of the response curve to osmolality; *hypervolemia* has an opposite effect, increasing the osmotic threshold and reducing the slope of the response curve (Fig. 56-1). Notably, AVP has a half-life in the circulation of only 10–20 min; thus, changes in ECF volume and/or circulating osmolality can rapidly affect water homeostasis. In addition to volume status, a number of other "nonosmotic" stimuli have potent activating effects on

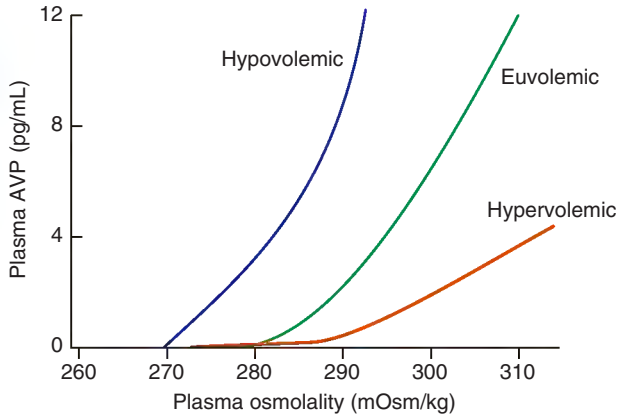

FIGURE 56-1 Circulating levels of vasopressin (AVP) in response to changes in osmolality. Plasma AVP becomes detectable in euvolemic, healthy individuals at a threshold of ~285 mOsm/kg, above which there is a linear relationship between osmolality and circulating AVP. The AVP response to osmolality is modulated strongly by volume status. The osmotic threshold is thus slightly lower in hypovolemia, with a steeper response curve; hypervolemia reduces the sensitivity of circulating AVP levels to osmolality.